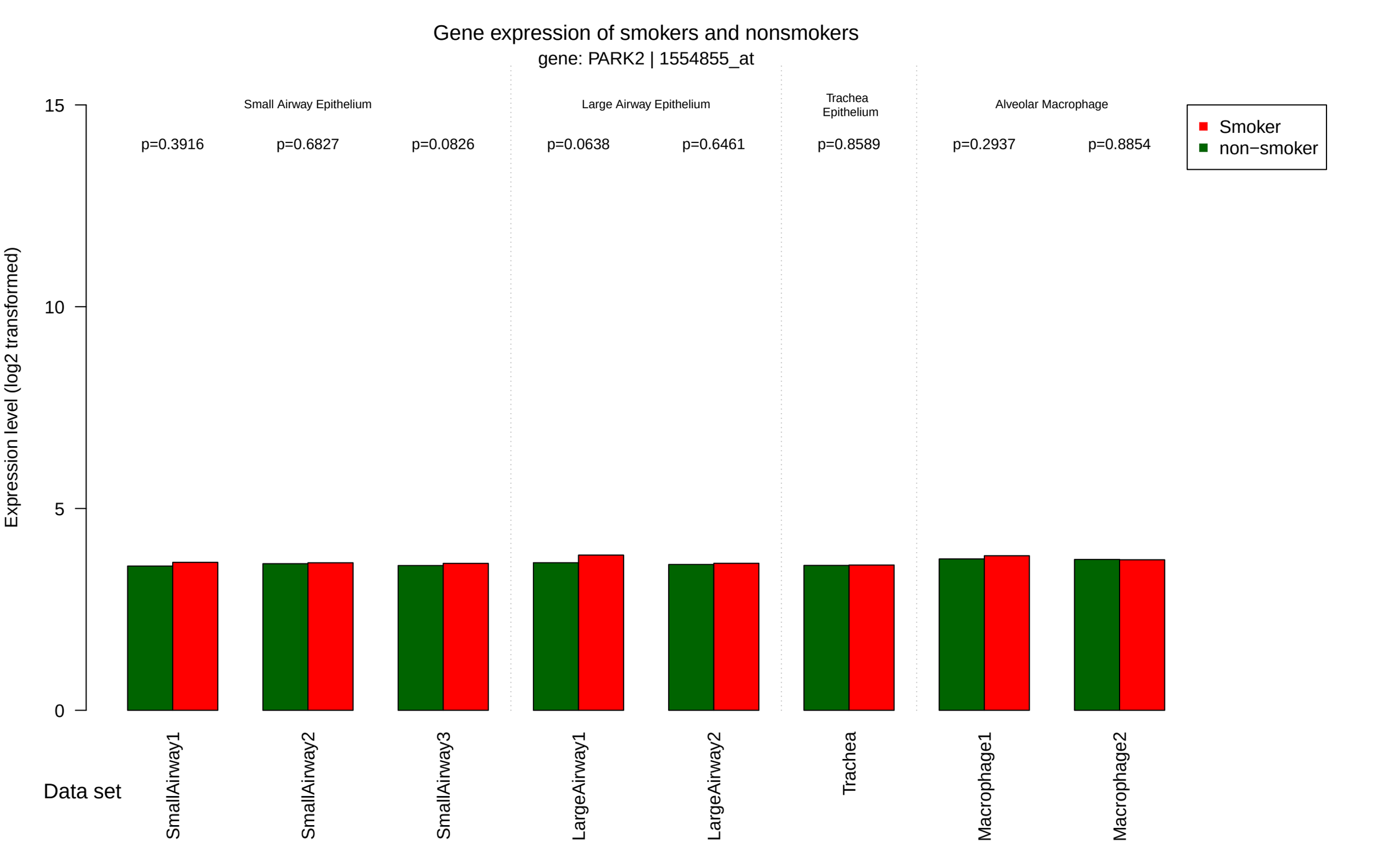
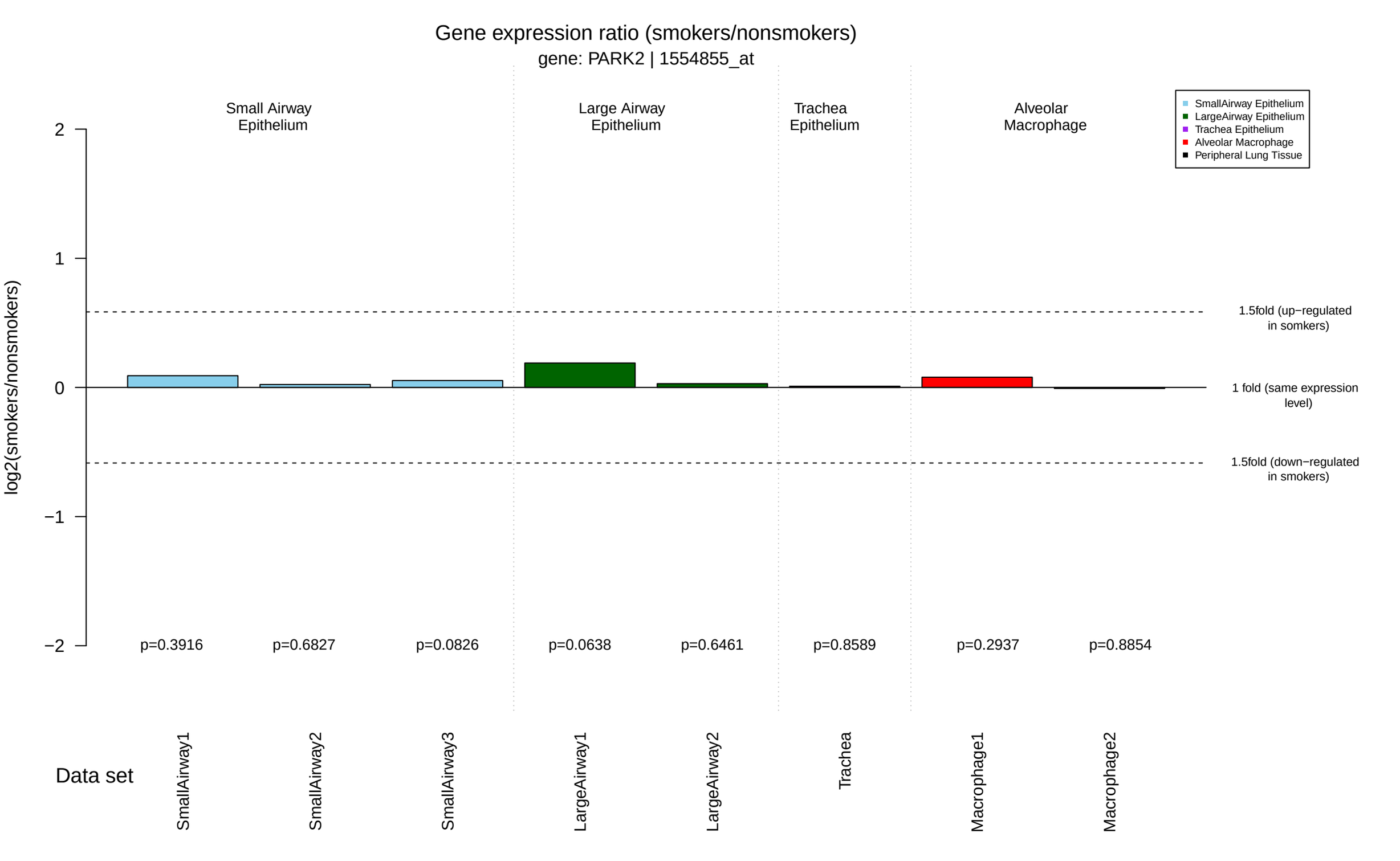
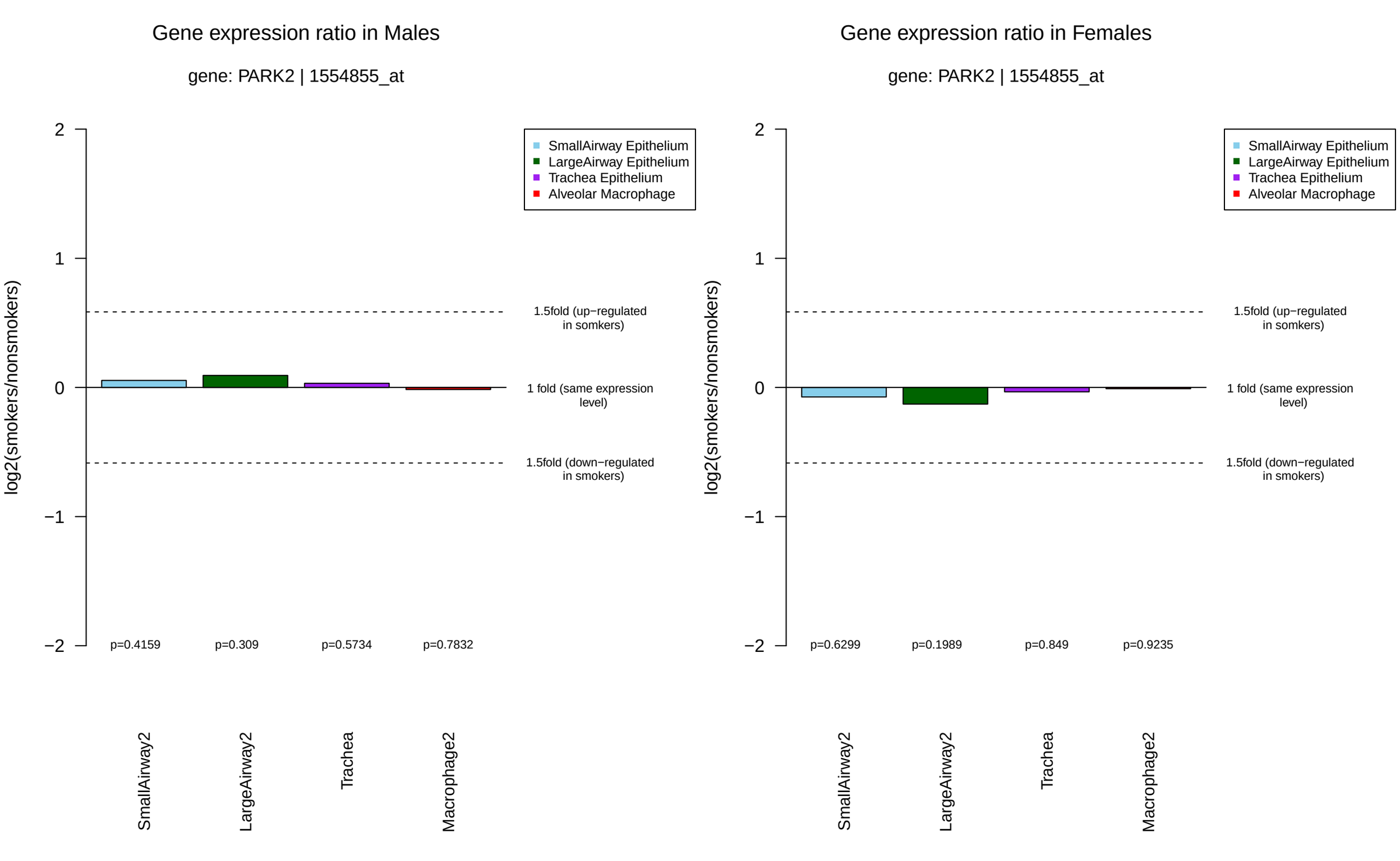
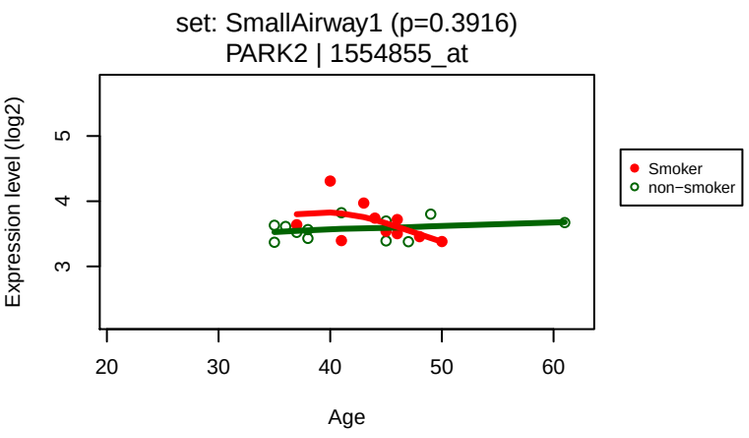
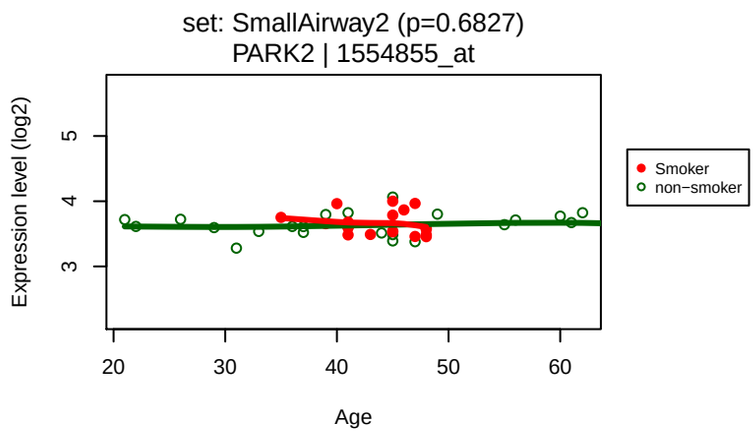
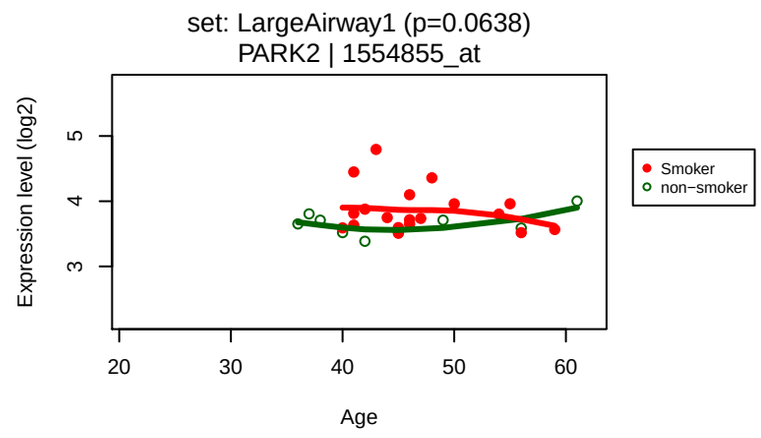

Gene expression of smokers and nonsmokers
gene: PARK2 | 1554855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3916
p=0.6827
p=0.0826
p=0.0638
p=0.6461
p=0.8589
p=0.2937
p=0.8854
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

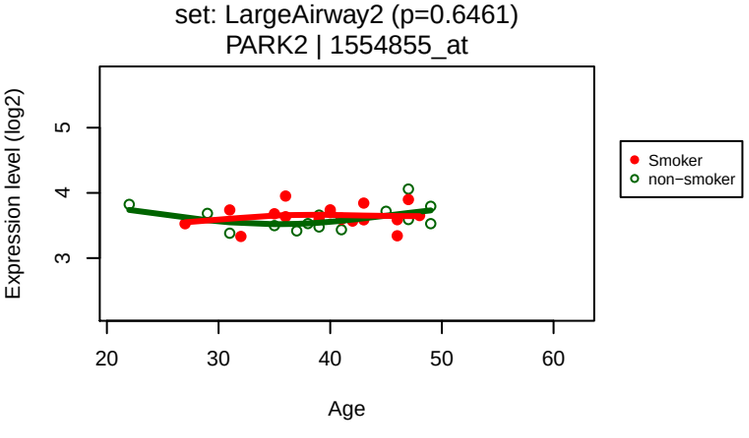

Gene expression ratio (smokers/nonsmokers)
gene: PARK2 | 1554855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3916
p=0.6827
p=0.0826
p=0.0638
p=0.6461
p=0.8589
p=0.2937
p=0.8854
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

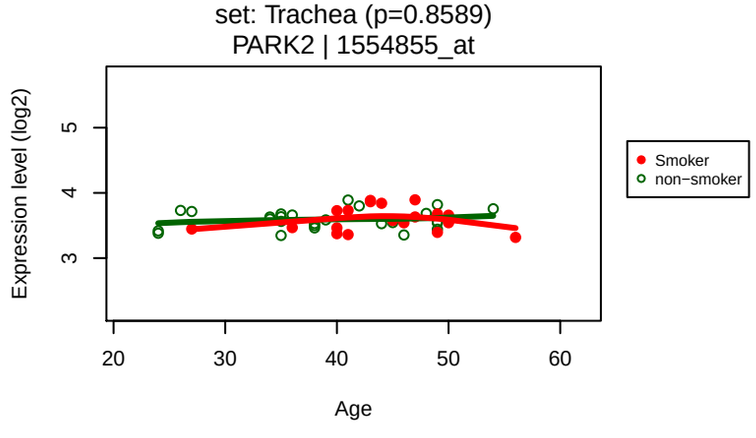

Gene expression ratio in Males
gene: PARK2 | 1554855_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4159
p=0.309
p=0.5734
p=0.7832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PARK2 | 1554855_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6299
p=0.1989
p=0.849
p=0.9235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

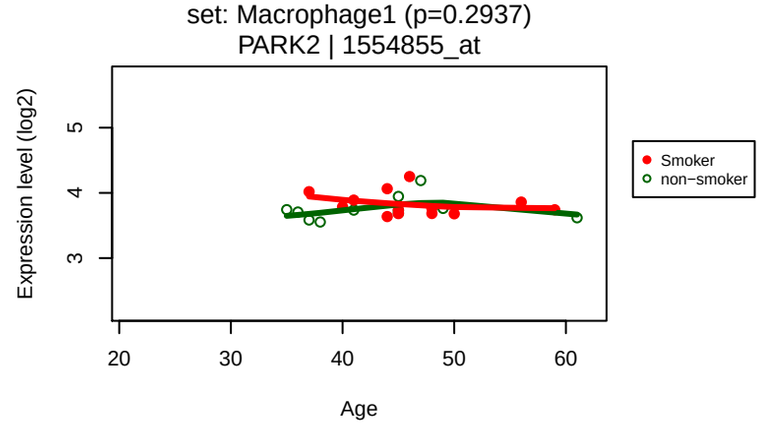

set: SmallAirway1 (p=0.3916)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker

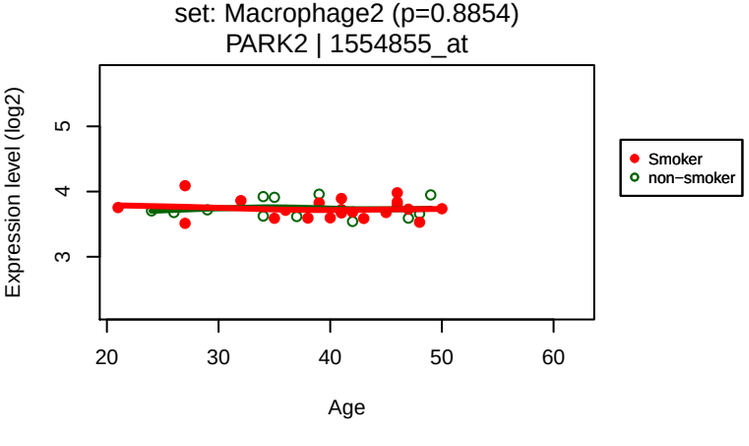

set: SmallAirway2 (p=0.6827)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker

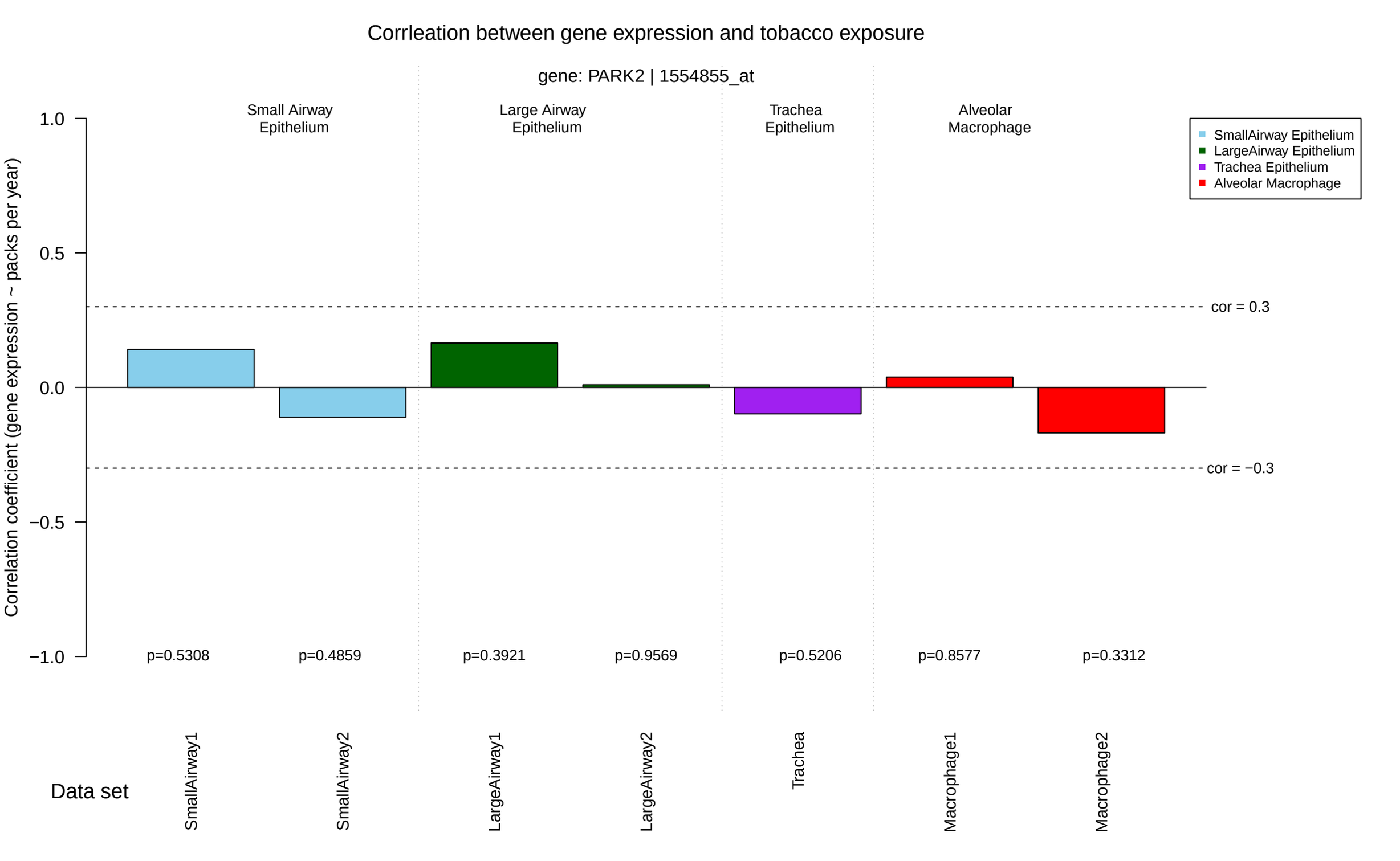

set: LargeAirway1 (p=0.0638)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6461)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8589)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2937)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8854)
PARK2 | 1554855_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PARK2 | 1554855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5308
p=0.4859
p=0.3921
p=0.9569
p=0.5206
p=0.8577
p=0.3312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2